

Activity 11

Writing a c program to perform the operation according to user's choice

1.finding the sum of the squares of digits of number user has input.

2.finding the sum of the cube of digits of number user has input.

3.finding the difference between cube and square sums of the digits

```

#include<stdio.h>
int main(){
    int num, choice;
    printf("Enter the number:");
    scanf("%d", &num);

    do {
        printf("\n-----Menu-----\n");
        printf("1.Sum of square of digits(while loop)");
        printf("2.Sum of cube of digits(for loop)");
        printf("Difference between cube and squares");
        printf("4.Exit the program\n");
        scanf("%d", &choice);

        switch(choice){
            case 1: {
                int temp = num, digit, sum = 0;
                while(temp > 0){
                    digit = temp % 10;
                    sum += digit * digit;
                    temp /= 10;
                }
                printf("Sum of square of digits:%d", sum);
                break;

            case 2: {
                int temp = num, sum = 0, digit;
                for(; temp>0; temp /= 10){
                    digit = temp % 10;
                    sum += digit * digit * digit;
                }
                printf("Sum of cubes:%d", sum);
                break;

            case 3: {
                int temp = num, digit, sumsq=0, sumcube=0;
                do{
                    digit = temp % 10;
                    sumsq += digit * digit;
                    sumcube += digit * digit * digit;
                    temp /= 10;
                }while(temp > 0);
                printf("difference of sum of squares and cube :%d\n", sumcube - sumsq);
                break;

            case 4: {
                printf("Exiting program.\n");
                break;

            default:
                break;
            }
        }
    }while(choice != 4);
    return 0;
}

```

```

        break;
    }
    default:
        printf("Invalid choice try again.\n");
}
}while(choice != 4);
return 0;

```

Compilation and execution

```

(kali@kali)-[~]
$ vi newd4.c

(kali@kali)-[~]
$ gcc newd4.c -o newd4

```

input	Expected OUTPUT	Actual OUTPUT	Result
12	5	5	PASS
13	10	10	PASS
17	50	50	PASS
19	82	82	PASS

(this test case if user select choice 1)

Input	Expected	Actual	Result
24	72	72	Pass
14	65	65	Pass
18	325	325	Pass

(for choice 2)

Input	Expected	Actual	Result
12	4	4	Pass

14	48	48	Pass
13	18	18	Pass

(for choice 3)

Algorithm

1.START

2. take input of number

3. use do while loop

4.ask choice of operation (1, 2, 3, 4)

5.Use switch , case keywords for each choice and operation for a user can select any.

6.use while loop in case1.

7. use for loop in case2.

8. use do while loop in case3.

9.use case 4 if user desires for exiting

10.STOP